

BACSE344 - Cloud Infrastructure And Architecture

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DA Category: DA1-Challenging Assignment

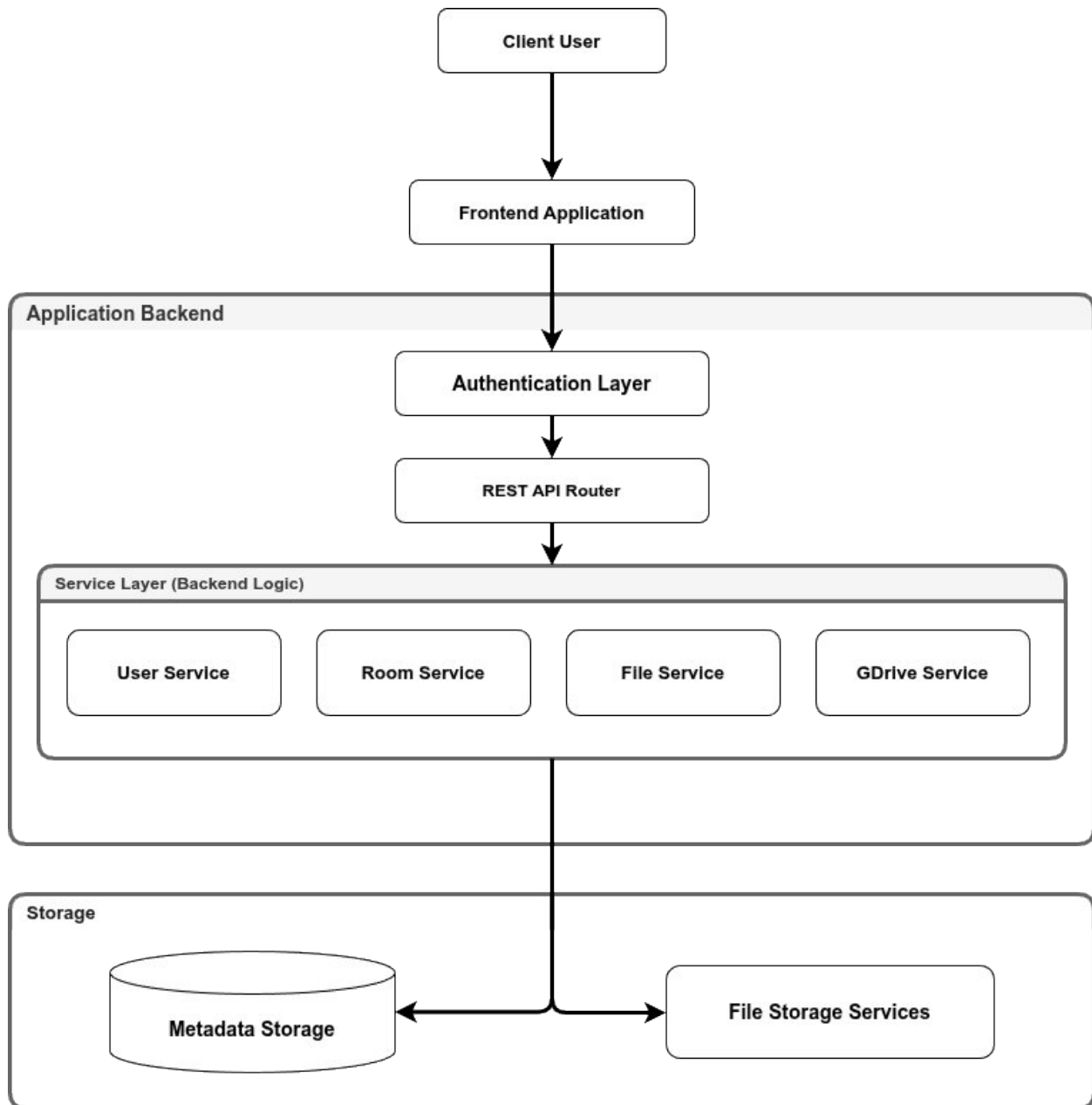
1.1 Problem Statement

For users operating within Google Drive's free tier, managing files across multiple accounts leads to storage fragmentation. While Google provides a 15 GB free storage allocation per account, relying on a single account creates capacity limits for project files, media, and documents. To avoid paid storage subscriptions, users frequently create secondary Google accounts or share folders across family, friends and teammates. However, this introduces account switching friction, fragmented storage management, and difficulty locating files as primary accounts hit capacity limits.

GatherAround aggregates scattered Google Drive storage quotas into a password protected virtual storage pool. Using a greedy bin packing algorithm, the platform dynamically routes file uploads to whichever linked Google Drive account has available capacity, presenting a single virtual file system hierarchy.

1.2 System Architecture & Component Diagram

The platform follows a layered, decoupled system architecture. Request processing flows sequentially from the client through an authentication boundary before reaching backend API routers, logic services, and storage components:



Component Responsibilities & System Interactions:

- 1. **Client & Frontend Application:**
 - Provides the single-page user interface for room management, quota sliders, and file navigation.
 - Captures user interactions and dispatches HTTP requests to the backend API.
- 2. **Authentication Layer:**
 - Validates JWT session tokens and enforces room-level access authorization before forwarding requests to backend services.
 - Rejects unauthorized or unauthenticated API requests at the entry boundary.
- 3. **Application Backend & Service Layer:**
 - **REST API Router:** Receives validated requests from the Authentication Layer and routes them to appropriate service handlers.
 - **Service Layer (Backend Logic):** Executes core application logic divided into four modules:
 - **User Service:** Manages user accounts, authentication tokens, and profile settings.
 - **Room Service:** Handles room creation, password validation, and member quota allocations.
 - **File Service:** Manages virtual folder hierarchies, file renaming, moving, and deletion.
 - **GDrive Service:** Manages OAuth 2.0 token refreshes and physical file streaming with the Google Drive API.
- 4. **Storage Container:**
 - **Metadata Storage:** Persists relational data in PostgreSQL, including user accounts, room membership allocations, password hashes, and virtual folder structures.
 - **File Storage Services:** Handles physical file storage, streaming downloads, and deletions on connected Google Drive accounts.

2. Application Development & Technology Stack

GatherAround is built using a client server architecture. The table below lists the primary technologies used for each component of the application:

Technology	Category	Usage in Application
React 18 & Vite	Frontend Framework	Single page web user interface for room management and file navigation
Vanilla CSS3	UI Styling	Responsive design system, custom sliders, and theme styling

Python 3.11 & FastAPI	Backend API Server	REST API endpoints, service logic, and request routing
Async SQLAlchemymy & asyncpg	Database ORM	Database modeling, async queries, and connection pooling
Neon PostgreSQL 16	Relational DBaaS	Data storage for users, rooms, membership quotas, and file metadata
Google Drive API v3	Cloud Storage API	Remote file uploads, folder creation, streaming, and deletions
Google OAuth 2.0 API	Identity & Authentication	User authentication and Google Drive access token management

3. Database Design

GatherAround uses a relational PostgreSQL database hosted on Neon DBaaS with four core tables:

Table	Description	Columns
users	User accounts & Google OAuth credentials	id, email, name, google_access_token, google_refresh_token, storage_limit, storage_usage
rooms	Virtual storage rooms created by users	id, name, password, owner_id
user_rooms	Room memberships & contributed storage allocations	user_id, room_id, allocated_bytes, used_bytes, files_hosted_count, gdrive_folder_id
file_items	Virtual folder hierarchies & file placement metadata	id, room_id, parent_id, name, is_folder, size_bytes, mime_type, uploader_id, storage_user_id, gdrive_file_id, created_at

4. REST API Routers

The backend architecture is divided into four primary API routers. For complete interactive testing, request/response schemas, and execution details, refer to the live [API Documentation](#):

Router Prefix	Scope & Responsibilities
/auth	User authentication & Google OAuth 2.0 token exchange
/users	User profiles & storage capacity details
/rooms	Room creation, password access, and member quota allocations
/files	Greedy bin-packed file uploads, virtual folder hierarchies, file streaming & deletion

4.1 CRUD & Database Operations Matrix

Operation	Endpoint	Method	Target DB Table & SQL Operation	Description
Create	/auth/signup	POST	INSERT INTO users	Registers new user account
	/rooms	POST	INSERT INTO rooms	Creates a new virtual storage room
	/rooms/{id}/join	POST	INSERT INTO user_rooms	Joins a room with password authentication
	/files/room/{id}/folder	POST	INSERT INTO file_items	Creates a new virtual folder
	/files/room/{id}/upload-intent-batch	POST	SELECT user_rooms (quota check)	Evaluates bin-packing placement for file uploads
	/files/room/{id}/complete-upload	POST	INSERT INTO file_items	Saves file record and links Google Drive asset ID
Read	/users/me	GET	SELECT FROM users	Retrieves profile &

				storage metrics
	/rooms	GET	SELECT FROM rooms JOIN user_rooms	Lists user accessible storage rooms
	/rooms/{id}/dashboard	GET	SELECT FROM user_rooms	Fetches storage pool breakdown & statistics
	/files/room/{id}	GET	SELECT FROM file_items	Retrieves virtual folder hierarchy
	/files/room/{id}/search	GET	SELECT FROM file_items WHERE name LIKE	Searches files by keyword
	/files/{id}/download	GET	SELECT FROM file_items + GDrive API	Streams file directly from Google Drive
Update	/rooms/{id}/contribute	POST	UPDATE user_rooms SET allocated_bytes	Updates member contributed storage quota
	/files/{id}/rename	PATCH	UPDATE file_items SET name	Renames a virtual file or folder
	/files/{id}/move	PATCH	UPDATE file_items SET parent_id	Relocates file/folder to new parent directory
Delete	/files/{id}	DELETE	DELETE FROM file_items	Deletes virtual file record and remote Drive asset
	/users/me	DELETE	DELETE FROM users (Cascade)	Deletes user profile & cleans up allocations

5. Cloud Deployment, Integration & Cloud Services Used

GatherAround deploys its system architecture components using three main cloud services: Platform as a Service (PaaS) for application hosting, Database as a Service (DBaaS) for data persistence, and Google Cloud APIs for drive integration:

Cloud Service	Provider & Service Category	Description
Render Static Site	Render PaaS (Frontend CDN)	Hosts and serves the web frontend interface.
Render Web Service	Render PaaS (Backend Web Container)	Runs the Python FastAPI container that processes API requests, user auth, and file routing logic.
Neon PostgreSQL	Neon DBaaS (Serverless Relational Database)	Persists user accounts, room settings, virtual folder hierarchies, and file metadata.
Google Cloud Console	Google Cloud Platform (Cloud Storage API)	Manages OAuth 2.0 credentials and provides API access to perform file uploads and storage management on Google Drive.

Section 6: Live Production Links & Source Code

- Deployed Application (Frontend): <https://gather-around.onrender.com>
- Deployed Backend API: <https://cloud-storage-aggregator.onrender.com>
- Interactive REST API Documentation: <https://cloud-storage-aggregator.onrender.com/docs>
- Source Code Repository: <https://github.com/mukesh-5059/Cloud-Storage-Aggregator>